AI Assisted Coding

Assignment 14.3

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Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contac information through a personal portfolio website. AI tools are used to speed u development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure wit sections:
   - Home
   - About
   - Projects
   - Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
   - Hover effects on project cards
   - Smooth scrolling navigation

Expected Outcome

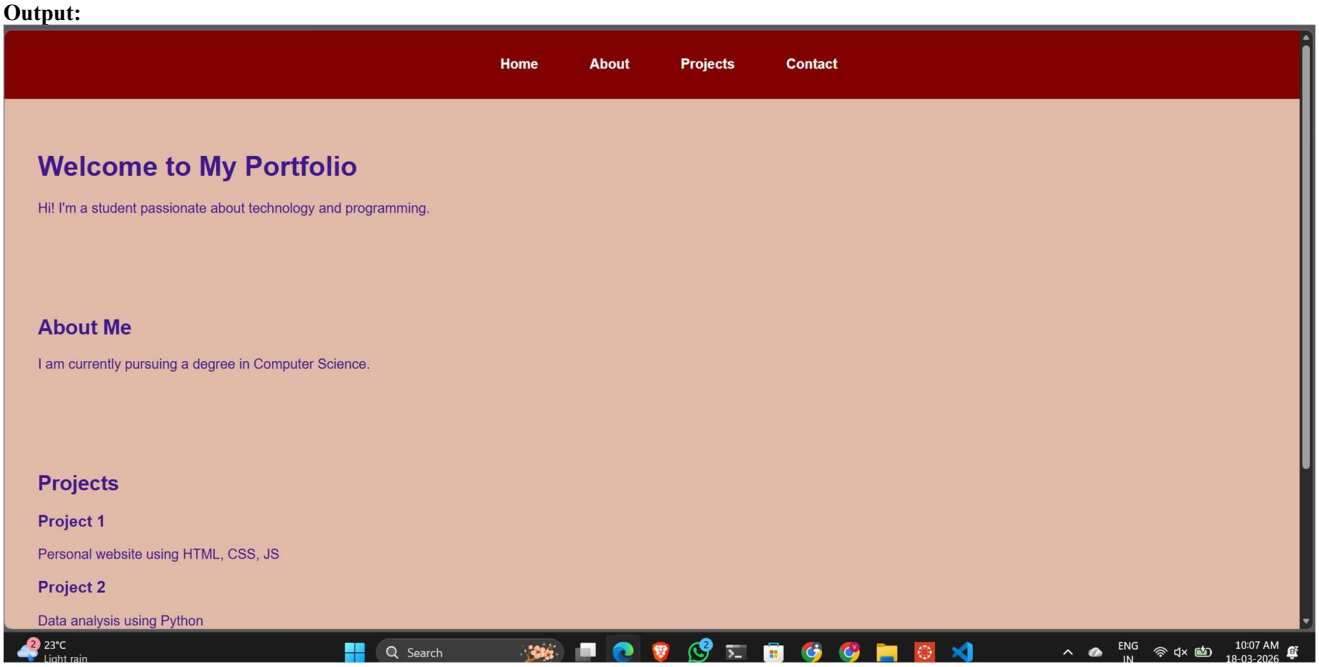
- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens
- Interactive project cards

Task 2: AI-Generated Restaurant Landing Page

CODE:

```html
<!--  Assume you are an experienced frontend developer building a modern personal portfolio website for a student.
 Your goal is to create a clean, professional, and responsive website that showcases academic projects, technical skills, and contact information.
HTML structure with
sections:
o Home
o About
o Projects
o Contact
- Add a fixed navigation bar with links to each section
Now improve the design to make it stand out from typical student portfolios.
Add a unique layout, better visual hierarchy, and modern UI/UX touches.
Avoid generic designs and make it look like a real production-level website.

 -->
<!-- <!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

    <nav class="navbar">
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#projects">Projects</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>

    <section id="home">
        <h1>Welcome to My Portfolio</h1>
        <p>Hi! I'm a student passionate about technology and programming.</p>
    </section>

    <section id="about">
        <h2>About Me</h2>
        <p>I am currently pursuing a degree in Computer Science.</p>
    </section>
```

Output:



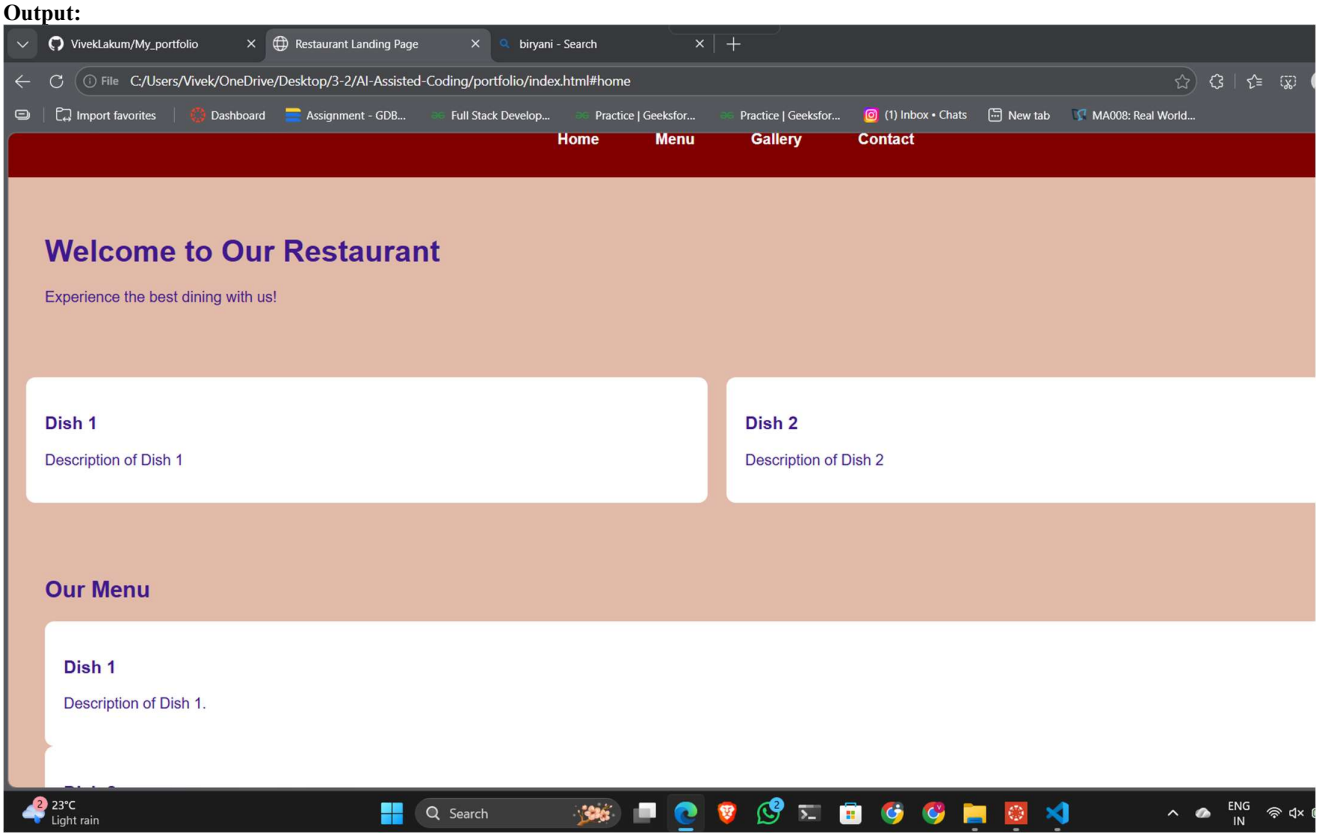
Task 2: AI-Generated Restaurant Landing Page
Scenario
A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.
Tasks
1. Use Copilot to create a navigation bar with links:
o Home
o Menu
o Gallery
o Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.
Expected Outcome
• Fully functional restaurant landing page
• Responsive navigation bar
• Working image slider with smooth transitions

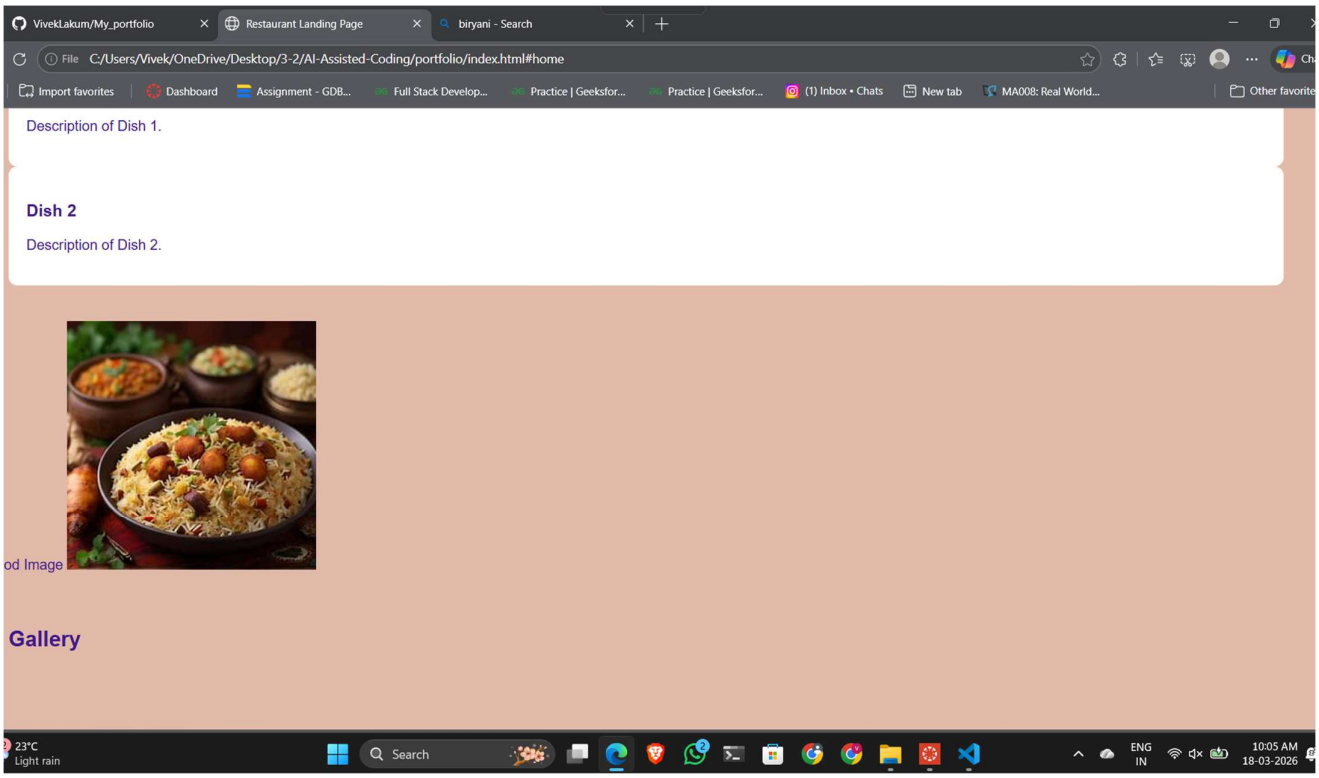
CODE:

```
</html>  -->
<!-- A local restaurant requires a simple yet attractive landing page to display its
offerings. The developer uses AI assistance to rapidly generate UI components.
Tasks
1. Use Copilot to create a navigation bar with links:
o Home
o Menu
o Gallery
o Contact
Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot
suggestions.
4. Customize animations or transitions for better user experience.
Expected Outcome
• Fully functional restaurant landing page
• Responsive navigation bar
• Working image slider with smooth transitions -->
<!-- <!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Landing Page</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <nav class="navbar">
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#menu">Menu</a></li>
            <li><a href="#gallery">Gallery</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>

    <section id="home">
        <h1>Welcome to Our Restaurant</h1>
        <p>Experience the best dining with us!</p>
    </section>
    <div class="menu-container">
  <div class="card">
```

```
<div class="card">
  <h3>Dish 1</h3>
  <p>Description of Dish 1</p>
</div>

<div class="card">
  <h3>Dish 2</h3>
  <p>Description of Dish 2</p>
</div>
/div>

  <section id="menu">
      <h2>Our Menu</h2>
      <div class="card">
          <h3>Dish 1</h3>
          <p>Description of Dish 1.</p>
      </div>
      <div class="card">
          <h3>Dish 2</h3>
          <p>Description of Dish 2.</p>
      </div>
  </section>
  <img src="" alt="Food Image">
  <img id="slide" src="OIP.jpg" alt="Gallery Image">
  <section id="gallery">
      <h2>Gallery</h2>
  </section>

  <section id="contact">
      <h2>Contact Us</h2>
      <p>Email: example@restaurant.com</p>
  </section>

/body>
/html> -->
```

Output:





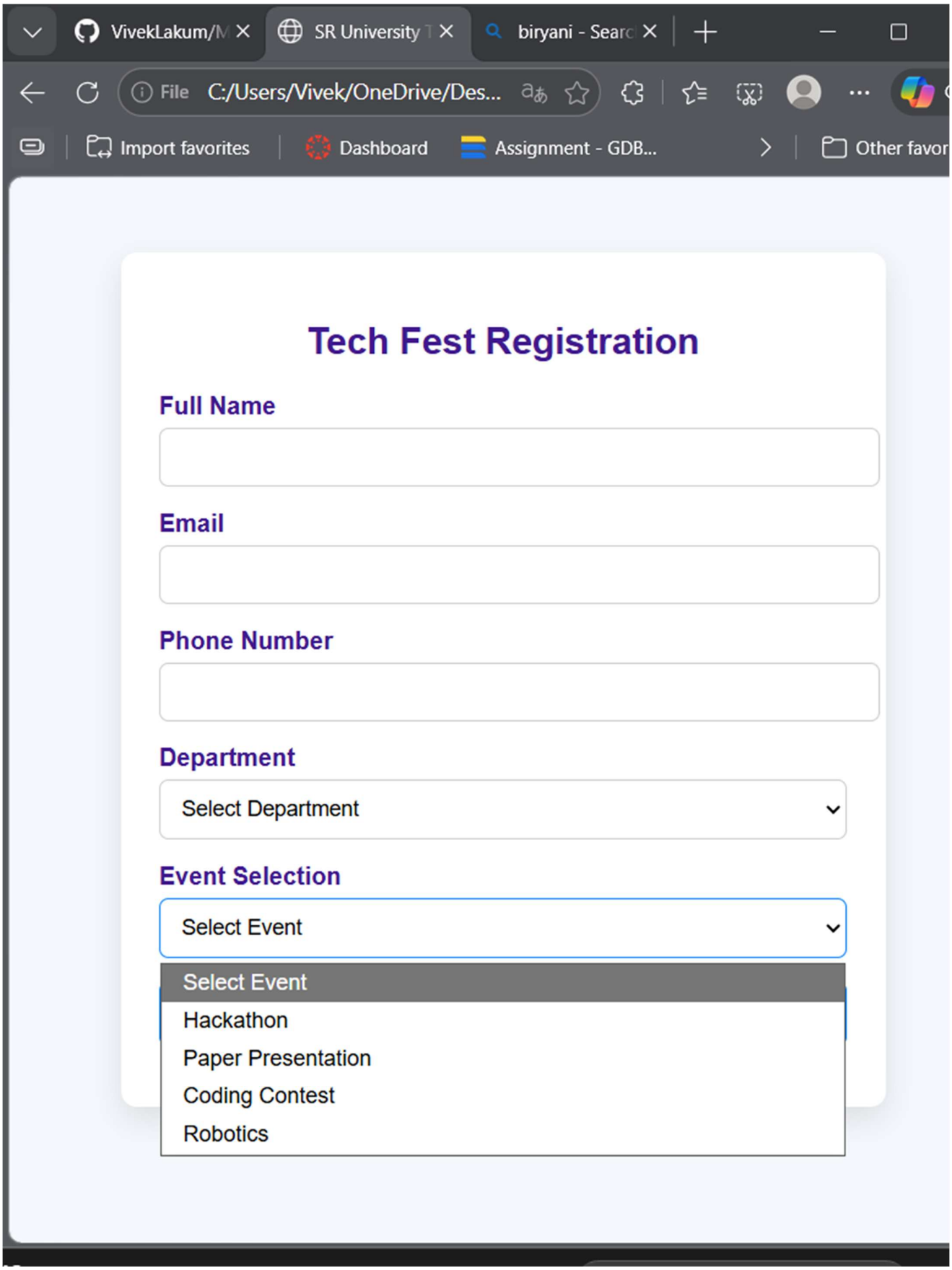
Task 3: AI-Powered Event Registration Form
Scenario
SR University is hosting a technical fest and requires an online registration formwith real-time validation to ensure accurate user input.
Tasks
1. Ask Copilot to generate an HTML form with the following fields:
o Name
o Email
o Phone Number
o Department
o Event Selection
2. Use AI assistance to style the form using CSS for readability and accessibility.
3. Implement JavaScript validation using Copilot:
o Email format validation
o Phone number length validation
o Required field checks
Expected Outcome
• User-friendly and visually appealing form
• Real-time validation messages
• Prevention of invalid submissions

CODE:

```
<!--  Create a complete responsive online registration form for "SR University Technical Fest" using HTML, CSS, and JavaScript.

Requirements:

1. HTML Structure:
- Include the following fields:
  • Full Name (text input)
  • Email (email input)
  • Phone Number (tel input)
  • Department (dropdown with options like CSE, ECE, MECH, CIVIL, etc.)
  • Event Selection (dropdown or checkbox list with multiple events)
- Add a submit button
- Use proper semantic HTML and labels for accessibility

2. CSS Styling:
- Make the form visually clean, modern, and centered on the page
- Use good spacing, readable fonts, and a soft color palette
- Add hover/focus effects for inputs and buttons
- Ensure responsiveness (mobile + desktop)
- Highlight errors clearly (red border + message)

3. JavaScript Validation (Real-time + On Submit):
- Validate email format using regex
- Validate phone number (must be exactly 10 digits)
- Ensure all required fields are filled
- Show real-time error messages below each field
- Prevent form submission if validation fails
- Display success message if all inputs are valid

4. Additional Requirements:
- Use only vanilla JavaScript (no frameworks)
- Organize code clearly (separate <style> and <script> sections or external files)
- Add comments for important logic
- Make sure UX is smooth (no page reload on validation errors)

Output:
- Provide full working code (HTML + CSS + JS in one file) -->
<!-- <!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>SR University Tech Fest Registration</title>
```

```
        }

        // Phone
        if (!validatePhone(phone.value)) {
            setError(phone, "Phone must be 10 digits", "phoneError");
            valid = false;
        } else {
            clearError(phone, "phoneError");
        }

        // Department
        if (department.value === "") {
            setError(department, "Select a department", "deptError");
            valid = false;
        } else {
            clearError(department, "deptError");
        }

        // Event
        if (event.value === "") {
            setError(event, "Select an event", "eventError");
            valid = false;
        } else {
            clearError(event, "eventError");
        }

        return valid;
    }

    // Real-time validation
    name.addEventListener("input", validateField);
    email.addEventListener("input", validateField);
    phone.addEventListener("input", validateField);
    department.addEventListener("change", validateField);
    event.addEventListener("change", validateField);

    form.addEventListener("submit", function(e) {
        e.preventDefault();

        if (validateField()) {
            document.getElementById("successMessage").innerText =
                "Registration successful!";
            form.reset();
        }
```

Output:

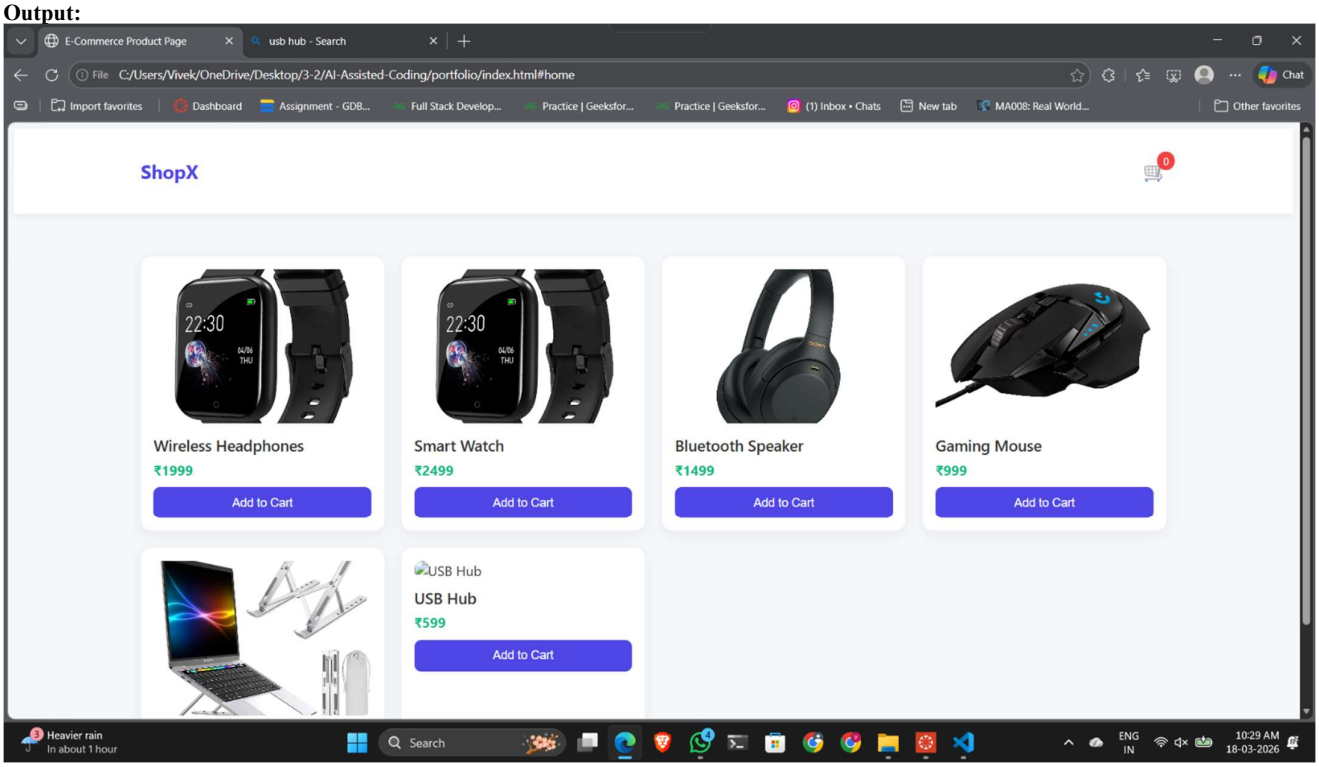

VivekLakum/M
SR University T
biryani - Searc
File
C:/Users/Vivek/OneDrive/Des...
Import favorites
Dashboard
Assignment - GDB...
Other favor
Tech Fest Registration
Full Name
Email
Phone Number
Department
Select Department
Event Selection
Select Event
Select Event
Hackathon
Paper Presentation
Coding Contest
Robotics

Task 4: AI-Assisted E-Commerce Product Page
Scenario
A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.
Tasks
1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an "Add to Cart" feature using JavaScript.
3. Modify AI-generated code to include:
o Cart item counter at the top-right corner
o Visual feedback when a product is added
Expected Outcome
• Functional product catalog
• Working cart counter
• Interactive user experience

```
<!--
Create a basic e-commerce product page using HTML, CSS, and vanilla JavaScript.

Requirements:

1. Product Catalog (HTML):
- Display products in a responsive grid layout
- Each product card must include:
  • Product image
  • Product name
  • Price
  • "Add to Cart" button
- Include at least 6 sample products

2. Header Section:
- Create a top navigation bar
- Add a cart icon at the top-right corner
- Display a cart item counter badge (initially 0)
Implement an "Add to Cart" feature using JavaScript.
3. Modify AI-generated code to include:
o Cart item counter at the top-right corner
o Visual feedback when a product is added


5. Code Structure:
- Link external CSS file (style.css)
- Write clean, readable code
- Use semantic HTML
- Keep JavaScript inside <script> tag or separate file

Output:
- Provide complete index.html
- Ensure it works without any external libraries
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>E-Commerce Product Page</title>

    <!-- Link CSS
    <link rel="stylesheet" href="style.css">
</head>
<body>
```

```html
<!-- ================ PRODUCTS ================
<div class="container">
    <div class="products">

        <!-- Product 1
        <div class="product-card">
            <img src="https://m.media-amazon.com/images/I/61ZjlBOp+rL._SL1500_.jpg" alt="Smart Watch">
            <div class="product-name">Wireless Headphones</div>
            <div class="price">₹1999</div>
            <button class="btn">Add to Cart</button>
        </div>

        <!-- Product 2 --
        <div class="product-card">
            <img src="https://m.media-amazon.com/images/I/61ZjlBOp+rL._SL1500_.jpg" alt="Smart Watch">
            <div class="product-name">Smart Watch</div>
            <div class="price">₹2499</div>
            <button class="btn">Add to Cart</button>
        </div>

        <!-- Product 3 --
        <div class="product-card">
            <img src="https://m.media-amazon.com/images/I/71o8Q5XJS5L._SL1500_.jpg" alt="Bluetooth Speaker">

            <div class="product-name">Bluetooth Speaker</div>
            <div class="price">₹1499</div>
            <button class="btn">Add to Cart</button>
        </div>

        <!-- Product 4 --
        <div class="product-card">
           <img src="https://m.media-amazon.com/images/I/61mpMH5TzkL._SL1500_.jpg" alt="Gaming Mouse">           <div class="product-na
            <div class="price">₹999</div>
            <button class="btn">Add to Cart</button>
        </div>

        <!-- Product 5 --
        <div class="product-card"><img src="https://rukminim2.flixcart.com/image/2400/2479/xif0q/shopsy-laptop-stand/l/x/y/200-adjustabl
            <div class="price">₹799</div>
            <button class="btn">Add to Cart</button>
        </div>

        <!-- Product 6 --
        <div class="product-card">
```

Output:



Task 5: Create a Responsive Web Page Layout
Instructions:
- **Design a basic web page layout with a header, main content area, and footer using HTML and CSS.**
- **Use AI to assist in generating responsive CSS for different screen sizes.**
- **Ensure the layout is clean and visually organized.**

Expected Output:
- **A responsive web page with:**
- **Header with navigation links**
- **Main section with placeholder text/images**
- **Footer with copyright or contact info**
- **Layout adapts correctly to desktop, tablet, and mobile screen sizes.**

CODE:

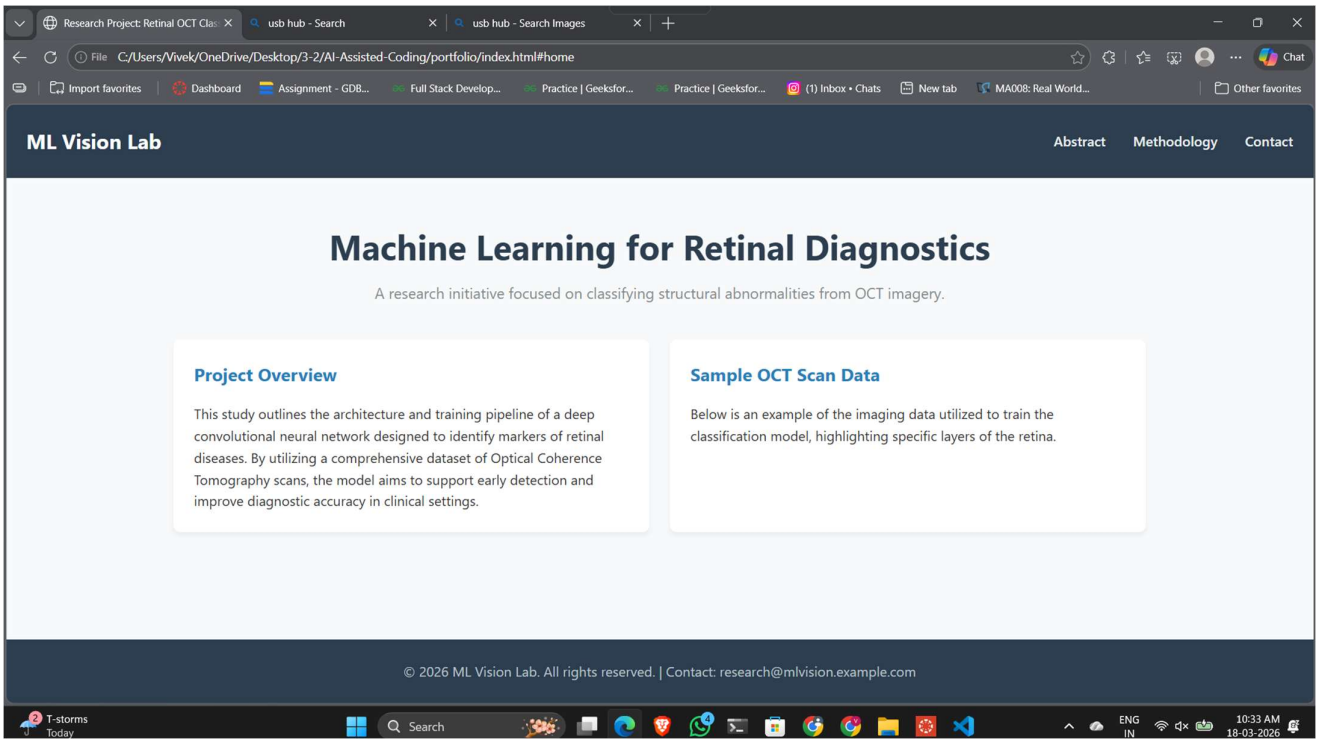
index.html > ...

```html
<html lang="en">
<body>
    <neaaer>
    </header>

    <main>
        <section class="hero">
            <h1>Machine Learning for Retinal Diagnostics</h1>
            <p>A research initiative focused on classifying structural abnormalities from OCT imagery.</p>
        </section>

        <section class="content-grid">
            <article class="card" id="abstract">
                <h2>Project Overview</h2>
                <p>This study outlines the architecture and training pipeline of a deep convolutional neural network designed to identify markers of retinal diseases. By utilizing a comp
            </article>

            <article class="card" id="methodology">
                <h2>Sample OCT Scan Data</h2>
                <p>Below is an example of the imaging data utilized to train the classification model, highlighting specific layers of the retina.</p>
                <div class="image-container">

                </div>
            </article>
        </section>
    </main>

    <footer>
        <p>&copy; 2026 ML Vision Lab. All rights reserved. | Contact: research@mlvision.example.com</p>
    </footer>

</body>
</html>
```

.
..
.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Research Project: Retinal OCT Classification</title>
    <style>
        /* ----------------------------------------
           BASE STYLES & MOBILE-FIRST LAYOUT
           ---------------------------------------- */
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            line-height: 1.6;
            color: #333;
            background-color: #f8f9fa;
            display: flex;
            flex-direction: column;
            min-height: 100vh; /* Ensures footer stays at the bottom */
        }

        /* Header & Navigation (Mobile Stacked) */
        header {
            background-color: #2c3e50;
            color: #ffffff;
            padding: 1.5rem;
            text-align: center;
        }

        .logo {
            font-size: 1.5rem;
            font-weight: bold;
            margin-bottom: 1rem;
        }

        nav {
            display: flex;
            flex-direction: column;
```

Output:


Research Project: Retinal OCT Clas
usb hub - Search
usb hub - Search Images
File C:/Users/Vivek/OneDrive/Desktop/3-2/AI-Assisted-Coding/portfolio/index.html#home
Chat
Import favorites
Dashboard
Assignment - GDB...
Full Stack Develop...
Practice | Geeksfor...
Practice | Geeksfor...
(1) Inbox • Chats
New tab
MA008: Real World...
Other favorites
ML Vision Lab
Abstract
Methodology
Contact
Machine Learning for Retinal Diagnostics
A research initiative focused on classifying structural abnormalities from OCT imagery.
Project Overview
This study outlines the architecture and training pipeline of a deep convolutional neural network designed to identify markers of retinal diseases. By utilizing a comprehensive dataset of Optical Coherence Tomography scans, the model aims to support early detection and improve diagnostic accuracy in clinical settings.
Sample OCT Scan Data
Below is an example of the imaging data utilized to train the classification model, highlighting specific layers of the retina.
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T-storms
Today
Search
ENG
IN
10:33 AM
18-03-2026